SUMMARY

1. For a given set of three vectors $\vec{a}, \vec{b}$ and $\vec{c}$, the scalar $(\vec{a}\times\vec{b})\cdot\vec{c}$ is called a **scalar triple product** of $\vec{a}, \vec{b}, \vec{c}$.
2. The **volume of the parallelepiped** formed by using the three vectors $\vec{a}, \vec{b}$, and $\vec{c}$ as co-terminus edges is given by $\left|(\vec{a}\times\vec{b})\cdot\vec{c}\right|$.
3. The scalar triple product of three non-zero vectors is zero if and only if the three vectors are **coplanar**.
4. Three vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar, if, and only if there exist scalars $r, s, t \in \mathbb{R}$ such that atleast one of them is non-zero and $r\vec{a} + s\vec{b} + t\vec{c} = \vec{0}$.
5. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{p}, \vec{q}, \vec{r}$ are any two systems of three vectors, and if $\vec{p} = x_1\vec{a} + y_1\vec{b} + z_1\vec{c}$, $\vec{q} = x_2\vec{a} + y_2\vec{b} + z_2\vec{c}$, and, $\vec{r} = x_3\vec{a} + y_3\vec{b} + z_3\vec{c}$, then $\left[\vec{p}, \vec{q}, \vec{r}\right] = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \left[\vec{a}, \vec{b}, \vec{c}\right]$.
6. For a given set of three vectors $\vec{a}, \vec{b}, \vec{c}$, the vector $\vec{a}\times(\vec{b}\times\vec{c})$ is called **vector triple product**.
7. For any three vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ we have $\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$.
8. Parametric form of the vector equation of a straight line that passes through a given point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r} = \vec{a} + t\vec{b}$, where $t \in \mathbb{R}$.
9. Cartesian equations of a straight line that passes through the point (x_1, y_1, z_1) and parallel to a vector with direction ratios b_1, b_2, b_3 are $\frac{x - x_1}{b_1} = \frac{y - y_1}{b_2} = \frac{z - z_1}{b_3}$.
10. Any point on the line $\frac{x - x_1}{b_1} = \frac{y - y_1}{b_2} = \frac{z - z_1}{b_3}$ is of the form $(x_1 + tb_1,\ y_1 + tb_2,\ z_1 + tb_3)$, $t \in \mathbb{R}$.
11. Parametric form of vector equation of a straight line that passes through two given points with position vectors $\vec{a}$ and $\vec{b}$ is $\vec{r} = \vec{a} + t\left(\vec{b} - \vec{a}\right), t \in \mathbb{R}$.
12. Cartesian equations of a line that passes through two given points (x_1, y_1, z_1) and (x_2, y_2, z_2) are $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$.
13. If θ is the acute angle between two straight lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$, then $\theta = \cos^{-1}\left(\frac{\left|\vec{b}\cdot\vec{d}\right|}{\left|\vec{b}\right|\left|\vec{d}\right|}\right)$
14. Two lines are said to be coplanar if they lie in the same plane.
15. Two lines in space are called **skew lines** if they are not parallel and do not intersect
16. The shortest distance between the two skew lines is the length of the line segment perpendicular to both the skew lines.
17. The shortest distance between the two skew lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ is $\delta = \frac{\left|(\vec{c} - \vec{a})\cdot(\vec{b}\times\vec{d})\right|}{\left|\vec{b}\times\vec{d}\right|}$, where $|\vec{b}\times\vec{d}| \neq 0$.

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